

Warm up: Skateboarding Sally

Sally is riding her skateboard down a straight road in one direction so that her distance x (in m) from her home is a function of the time t (in s) since she left home. We know that **60 seconds after leaving home, she has travelled 150 metres and her current velocity is 5m/s.**

Poll Question

- (a) About how much further will Sally have travelled after an additional 2 seconds has passed? About how far from home will she be overall?
- (b) About how far from home will Sally be after 120 seconds has passed?
- (c) Which of our two estimates do you have the most confidence in?



Part (a)



Part (b)

Lecture 9

Linear Approximation The Second Derivative

Big Questions

- How can we estimate unknown function values based on data?
- How can we describe the rate at which an object's velocity is changing?

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Poll Question

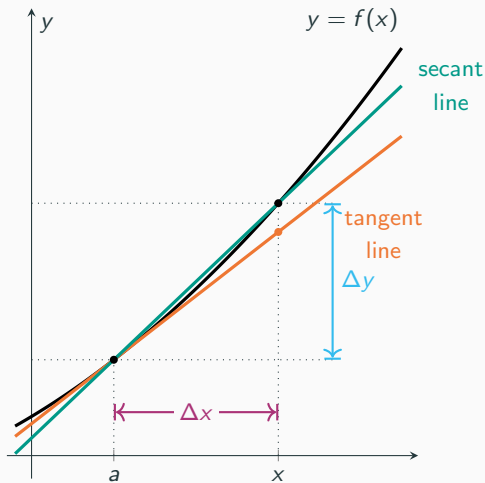
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A

Part (a)

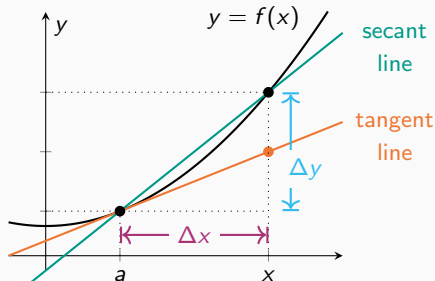
B

Part (b)



Tangent Line Approximation (a.k.a Linear Approximation)

Suppose that we know $f(a)$ and $f'(a)$, and we want to estimate $f(x)$.



As long as $\Delta x = x - a$ is small,

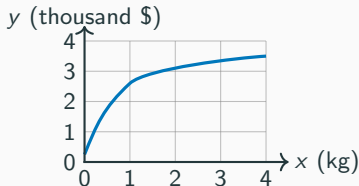
$$\frac{\Delta y}{\Delta x} \approx \boxed{} \quad \text{and so } \Delta y \approx \boxed{}$$

Therefore

$$f(x) \approx$$

Estimating a nearby value

Suppose that the cost $y = f(x)$, in thousands of dollars, of manufacturing x kilograms of ibuprofen is shown on the right.



Poll Question

- (a) It is known that $f(1) = 2.6$ and $f'(1) = 0.6$. Estimate the cost of producing 1.15 kg of ibuprofen.
- (b) Is your estimate an **A** overestimate or **B** underestimate? Why?
- (c) Calculate $\frac{f'(1)}{f(1)}$ and interpret its meaning.

Relative Rate of Change

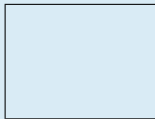
In addition to average rates of change and instantaneous rates of change, we have a notion of *relative rates of change*.

Definition

Let $Q = f(t)$. The **relative rate of change** of Q with respect to t , at $t = a$ is defined as

relative rate of change
of Q at a

=



The relative rate of change expresses the instantaneous rate of change as a percentage of .

Interpreting relative rates of change

Discussion Question

The area of Brazil's rainforest $R = f(t)$, in million acres, is a function of the number of years t since 2000. According www.rain-tree.com, $f(9) = 740$ and $f'(9) = -2.7$.

- (a) Find the relative rate of change of $f(t)$ when $t = 9$.

- (b) Interpret your result from (a) using non-technical language accessible to the general public.

Second Derivatives: A Classic Example

Discussion Question

Suppose that $x = f(t)$ is the position of a car (in km) along a straight road at time t (in hours) from a fixed location. Fill in the blanks with the appropriate physical quantities:

Position (x)

$\frac{d}{dt} \rightarrow$
derivative

$\frac{d}{dt} \rightarrow$
derivative

From this, we see that acceleration is the

of position, i.e. $a =$

Distance, velocity and acceleration (part 1)

Poll Question

A car drives through an intersection after the stop light changes from red to green. If $f(t)$ is the distance travelled by the car t seconds after it takes off, $0 \leq t \leq 3$, then,

- | | | | |
|--------------------|---|--|--|
| (a) $f'(t)$ is... | ☒ A positive | ☐ B negative | ☐ C zero |
| (b) $f''(t)$ is... | ☒ A positive | ☐ B negative | ☐ C zero |

Distance, velocity and acceleration (part 2)

Poll Question

Suppose $f(t)$ is the distance travelled by a person skating along a very slippery ice rink at constant velocity, $0 \leq t \leq 5$. Then,

- | | | | |
|--------------------|--|---|-------------------------------------|
| (a) $f'(t)$ is... | ☒ A positive | ☐ B negative | ☐ C zero |
| (b) $f''(t)$ is... | ☒ A positive | ☐ B negative | ☐ C zero |

Distance, velocity and acceleration (part 3)

Poll Question

Suppose $f(t)$ is the height of a coconut falling from a Palm tree, $0 \leq t \leq 1.8$.
Then

(a) $f'(t)$ is...

A

positive

B

negative

C

zero

(b) $f''(t)$ is...

A

positive

B

negative

C

zero

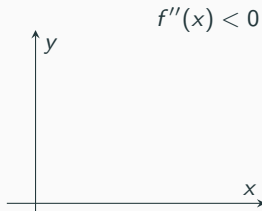
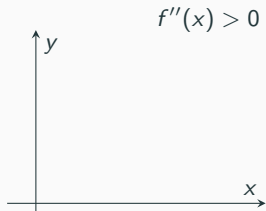
What can a function's second derivative tell you about its graph?

Poll Question

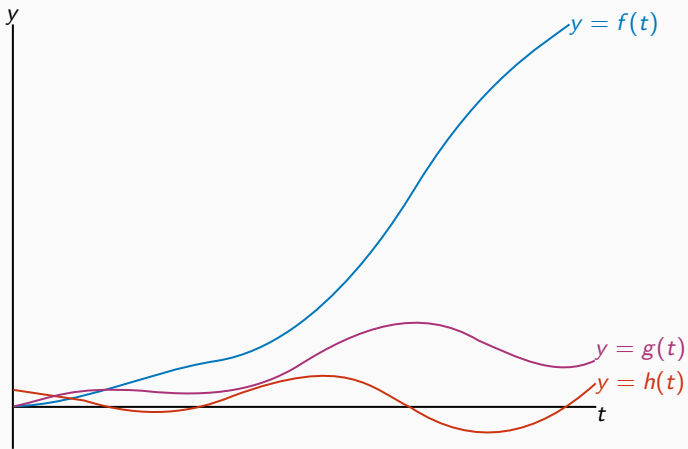
(a) **Suppose that $f''(x) > 0$ on an interval.** This means that, on that interval ☐ **A** f ☐ **B** f' ☐ **C** f'' is increasing. Hence, as we move to the right along the graph of $y = f(x)$, the ☐ **A** graph ☐ **B** slope is increasing. Therefore, the graph of $y = f(x)$ is ☐ **A** concave up ☐ **B** concave down.

(b) **Suppose that $f''(x) < 0$ on an interval.** By similar reasoning, this means that the graph of $y = f(x)$ is on that interval.

Sketch two pictures below to illustrate your answers.



The graphs of distance, velocity and acceleration of a moving car as functions of time t are shown below.



Discussion Question

Which is the graph of distance vs. time? Velocity? Acceleration?

- Complete questions 1–3 from the Week 3 Checkpoint Workbook.
- Homework 2 and the Reflection Questions are due tomorrow at 10pm.
- Please read Section 2.4.
- Complete #1–10 in Homework 3.
- The **individual part** and **pod contract** for Project 1 are due tomorrow at 10pm!

Have a great day!

Extra Practice: Four Curves

In each box, sketch a graph that satisfies the relevant conditions.

	$f''(x) > 0$	$f''(x) < 0$
$f'(x) > 0$		
$f'(x) < 0$		

Looking ahead: Estimating the sign of a second derivative from a table

The function w is defined by the table of values below. Does w' appear to be positive or negative? Does w'' appear to be positive or negative?

t	100	110	120	130	140
$w(t)$	10.7	6.3	4.2	3.5	3.3

- A** $w' > 0$ and $w'' > 0$
- B** $w' > 0$ and $w'' < 0$
- C** $w' < 0$ and $w'' > 0$
- D** $w' < 0$ and $w'' < 0$
- E** None of the above